

Internship Report – Day 16

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 16 Overview:

I have tested the model today with the modified python code which exactly takes and loads the model I have trained.

Just tried the new yolov8s.pt model how it going to work that's it .

Tried the new code in python for the accuracy.

while testing I have noticed that the model is not working properly so I have implemented some changes in that python code.

while testing I have even modified about the IND letter printing in the terminal I have modified the code to ignore that thing and print the numbers in a correct format.

Tools Used Today

GEMINI, SONNET, Chatgpt, vs coder, cmd terminal,trae.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

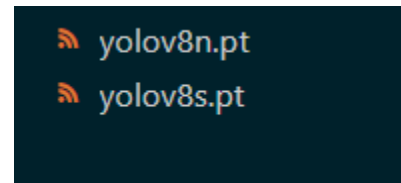
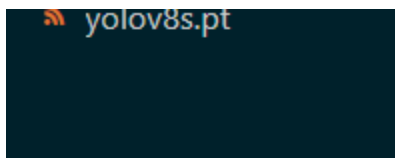
Approaches:-

I have used the yolov8n model and yolov8s models in project . And I have also used the utils for the clarification of the numbers and characters in the numberplate .

Taken the help form chatgpt to sole the command line issue.

And also solved the issue of installing the python 3.13.3 version to run the model perfectly.

Attachments:-



Dataset

▼ detected_plates

-  plate_20250519_102722_KA19.jpg
-  plate_20250519_102725_KA19EQ0001.jpg
-  plate_20250519_102728_KA19EQ0001.jpg
-  plate_20250519_103339_KA19EQ0001.jpg
-  plate_20250519_103341_KA19EQ0001.jpg
-  plate_20250519_103345_KA19EQ0001.jpg
-  plate_20250519_103401_6425.jpg
-  plate_20250519_103403_6425.jpg
-  plate_20250519_103405_06X.jpg
-  plate_20250519_103407_6425.jpg
-  plate_20250519_103409_6L25.jpg
-  plate_20250519_103427_RJIZCVOOOZ.jpg
-  plate_20250519_103429_RJIZCVO002.jpg
-  plate_20250519_103439_RJIZCVQ002.jpg
-  plate_20250519_103441_RJIZCVQ002.jpg
-  plate_20250519_103443_RJILCV0002.jpg
-  plate_20250519_103445_RJILCVOO02.jpg
-  plate_20250519_103447_RJILCVOOO2.jpg
-  plate_20250519_103540_Q0001.jpg
-  plate_20250519_103541_00001.jpg
-  plate_20250519_103543_Q0001.jpg
-  plate_20250519_103544_Q0001.jpg
-  plate_20250519_103546_Q0001.jpg

```

new.py > ...
1 import cv2
2 import easyocr
3 import numpy as np
4 import time
5 import os
6 import sys
7 import datetime
8 import re
9
10 # Configuration
11 CAMERA_ID = 0
12 FRAME_WIDTH = 640
13 FRAME_HEIGHT = 480
14 SAVE_PATH = "detected_plates"
15
16 # Initialize EasyOCR
17 print("Initializing EasyOCR...")
18 reader = easyocr.Reader(['en'], gpu=False)
19 print("EasyOCR initialized")
20
21 # Load prebooked plates
22 with open("prebooked_plates.txt", "r") as f:
23     prebooked = set(line.strip() for line in f if line.strip())
24     print(f"Loaded {len(prebooked)} prebooked plates")
25
26 # Create directory for saving plates
27 if not os.path.exists(SAVE_PATH):
28     os.makedirs(SAVE_PATH)
29     print(f"Created directory: {SAVE_PATH}")
30
31 def is_valid_plate_format(text):
32     """Check if the text matches common Indian license plate formats"""
33     # Remove any spaces and convert to uppercase
34     text = text.replace(" ", "").upper()
35
36     # Common Indian license plate patterns:
37     # 1. KA19EQ0001 (2 letters, 2 numbers, 2 letters, 4 numbers)
38     # 2. KA01AB1234 (2 letters, 2 numbers, 2 letters, 4 numbers)
39     # 3. KA01A1234 (2 letters, 2 numbers, 1 letter, 4 numbers)
40     # 4. KA01AB123 (2 letters, 2 numbers, 2 letters, 3 numbers)
41     patterns = [
42         r"[A-Z]{2}\d{2}[A-Z]{2}\d{4}$", # KA19EQ0001
43         r"[A-Z]{2}\d{2}[A-Z]{2}\d{4}$", # KA01AB1234
44         r"[A-Z]{2}\d{2}[A-Z]\d{4}$", # KA01A1234
45         r"[A-Z]{2}\d{2}[A-Z]{2}\d{3}$", # KA01AB123
46     ]
47
48     return any(bool(re.match(pattern, text)) for pattern in patterns)
49
50 def detect_and_recognize_plates():
51     """Detect and recognize license plates using contour-based detection"""
52     # Initialize webcam
53     cap = cv2.VideoCapture(CAMERA_ID)
54     cap.set(cv2.CAP_PROP_FRAME_WIDTH, FRAME_WIDTH)
55     cap.set(cv2.CAP_PROP_FRAME_HEIGHT, FRAME_HEIGHT)
56
57     if not cap.isOpened():
58         print("Error: Could not open camera")
59         return

```

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